

Task 1.

Tech "Two techniques are applied by each subject: Inspection and Testing."

- Unit: Inspection and Testing
- Scale type: Nominal
- Objective/Subjective: Objective
- Direct/indirect: Direct

Prog "The objects are two c programs from the PSP course"

- Unit: Counter and correlation
- Scale type: Nominal
- Objective/Subjective: Objective
- Direct/indirect: Direct

Exp" The experience with inspection and testing is measured on a four-level ordinal scale"

- Unit: Ordinal
- Scale type: Ordinal (1-4)
- Objective/Subjective: Objective (pre-test questionnaire)
- Direct/indirect: Indirect

Skill" The experience with inspection and testing is measured on a four-level ordinal scale."

- Unit: Ordinal
- Scale type: Ordinal (grading of two home-work tasks)
- Objective/Subjective: Subjective
- Direct/indirect: Indirect

Time" The time spent by each subject in performing the task. The time unit used is minutes."

- Unit: Minutes (Integer)
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

DEFI (The number of defects detected by each reviewer is recorded, excluding false positives.)

- Unit: integer
- Scale type: ratio
- Objective/Subjective: objective
- Direct/indirect: direct

DEFD (The number of defects isolated by each reviewer is recorded, excluding false positives.)

- Unit: integer
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

EFFI (Number of defects isolated per hour)

- Unit: Defects isolated per hour
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Indirect

EFFD (number of defects detected per hour)

- Unit: defects detected per hour
- Scale type: Ratio
- Objective/Subjective: objective
- Direct/indirect: Indirect

FNDI (The number of subjects isolate a certain defect in a specific program)

- Unit: Integer
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

FNDD (The number of subjects detect a certain defect in a specific program)

- Unit: Integer
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

ISOL" The ratio of the subjects that isolated the defect if it was detected."

- Unit: Ratio of the subjects that isolated the defect if it was detected (fraction)
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Task 2.

Defect Density (defects per lines of code)

- Unit: defects per lines of code (fraction)
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Student year (measure which year the students are)

- Unit: Integer
- Scale type: Ordinal
- Objective/Subjective: Objective
- Direct/indirect: Direct

Readability scoring (how readable a code snippet is)

- Unit: Integer (1 to 5)
- Scale type: Ordinal
- Objective/Subjective: Subjective
- Direct/indirect: Direct

Cohen's k "Measure of inter-rater agreement for categorical items"

- Unit: Decimal
- Scale type: Ordinal (-1 to 1)
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Linear weighted k "conceptually gives some credit for rankings that are "close""

- Unit: Decimal
- Scale type: Ordinal (-1 to 1)
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Kendall's t "number of bubble-sort operations to order one list in the same way as a second"

- Unit: Decimal
- Scale type: Ordinal (-1 to 1)
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Person's r "measures the degree of linear dependence"

- Unit: Decimal
- Scale type: Interval (-1 to 1)
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Spearman's ρ “(the degree of dependence with any arbitrary monotonic function”

- Unit: Decimal
- Scale type: Ordinal (-1 to 1)
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Cyclomatic complexity (number of independent paths)

- Unit: Integer
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

Method length (number of lines of code)

- Unit: Number of lines of code (Integer)
- Scale type: Ratio
- Objective/Subjective: Objective
- Direct/indirect: Direct

Maturity (What stage of development the software project is at)

- Unit: number for category
- Scale type: Ordinal
- Objective/Subjective: Subjective
- Direct/indirect: Indirect

Defects detected by FindBugs (did the function have a defect)

- Unit: Binary
- Scale type: Nominal
- Objective/Subjective: Objective
- Direct/indirect: Indirect

Defects detected by developers (did the function have a defect)

- Unit: Binary
- Scale type: Nominal
- Objective/Subjective: Subjective
- Direct/indirect: Indirect

Code churn (Measures if there was a change in the code)

- Unit: Binary
- Scale type: Nominal
- Objective/Subjective: Objective
- Direct/indirect: Direct